The logarithm is necessary: exact critical scaling of the uniform doubling condition on the ball

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June 2026

Abstract

In two companion papers we proved that N independent uniform samples from a convex body satisfy the uniform doubling condition of [Shmalo, *Ann. Math. Artif. Intell.* (2024)] with high probability, and that for untruncated slabs over a body with an interior rolling ball the scaling factor can be taken $\ell_N = O\big((\log N/N)^{2/(n+1)}\big)$, while single gap configurations force $\ell_N \geq cN^{-2/(n+1)}$ with probability bounded below. Two questions remained: whether the failure probability can be driven to 1, and whether the logarithm is an artifact of the Vapnik–Chervonenkis union bound. We answer both for the unit ball, $n \geq 2$: with probability tending to one, among $\asymp (N/\log N)^{(n-1)/(n+1)}$ disjoint boundary caps at least one contains a single, isolated, extremal sample point whose surrounding slabs stall at empirical mass $1/N$ through a full grid step at width scale $(\varepsilon \log N/N)^{2/(n+1)}$. Consequently the uniform doubling condition for pure slabs fails, with probability $1 - o(1)$, for every scaling factor $\ell \leq c(\kappa)\,(\log N/N)^{2/(n+1)}$. Combined with the companion upper bound, the critical scaling of the uniform doubling condition on the ball is

$$\ell_N \;\asymp\; \Big(\frac{\log N}{N}\Big)^{2/(n+1)},$$

exactly, logarithm included. The proof is a second-moment argument over the cap family; the only geometric input is the two-sided spherical cap-volume asymptotic.

1 Introduction

Let $K = B^n \subset \mathbb{R}^n$ be the closed unit ball, $n \geq 2$, let μ be the uniform probability measure on K, and let $T = \{X_1, \dots, X_N\}$ be i.i.d. samples with empirical measure $\bar{\mu}_N$. For a unit vector u, a center s, and a width w, write $S_w = S_w(u, s) = \{x \in \mathbb{R}^n : |\langle x - s, u\rangle| \leq w/2\}$ for the (pure, untruncated) slab. The uniform doubling condition (UDC) of [2] with constants $(\kappa, \delta, \sigma, \beta, \ell)$ requires

$$\bar{\mu}_N(S_{\ell\kappa^{i+1}}) \;\geq\; \min\big\{\delta,\; (1+\sigma)\,\bar{\mu}_N(S_{\ell\kappa^i})\big\}$$

for every slab and every $i \in \mathbb{N}_0$ with $\ell\kappa^i < \beta$; it is the hypothesis of the stability-of-accuracy theorems of [1, 2]. In [3] we showed that i.i.d. samples satisfy the UDC with probability $1 - \frac{1}{N}$ and scaling factor $\ell_N = O((\log N/N)^{1/n})$, for all uniformly thick truncated-slab families at once; in [4] we showed that for *pure* slabs over a body with an interior rolling ball the scaling factor improves to

$$\ell_N \;=\; O\Big(\Big(\frac{\log N}{N}\Big)^{2/(n+1)}\Big),$$

that this exponent is dictated by the spherical cap volume, and that truncations genuinely shift the critical exponent to $\frac{1}{n}$. The lower-bound constructions in both papers produce a *single* violating configuration, hence only a failure probability bounded below by a constant, and at scale $N^{-2/(n+1)}$ — without the logarithm. This left [4, Problem 5.1]:

> *Does the UDC for pure slabs on the ball hold at scaling $\ell \asymp N^{-2/(n+1)}$, or is the logarithmic factor necessary?*

We prove that the logarithm is necessary, and that the failure below the logarithmic scale is overwhelming rather than merely non-negligible.

Theorem 1.1. *Let $n \geq 2$, let μ be uniform on the unit ball, and fix $\kappa > 1$, $\sigma > 0$, $\beta > 0$, and $\delta \geq \frac{3}{N}$. There are constants $\varepsilon_0 = \varepsilon_0(n,\kappa) > 0$ and $c = c(n,\kappa) > 0$ such that, with probability $1 - o(1)$ as $N \to \infty$, the doubling inequality fails for some pure slab at every grid scale below $c\,(\log N/N)^{2/(n+1)}$. Consequently, with probability $1 - o(1)$, the sample satisfies the UDC for pure slabs with* no *scaling factor*

$$\ell \;\leq\; c\Big(\frac{\log N}{N}\Big)^{2/(n+1)}.$$

Together with the upper bound of [4], the critical scaling of the uniform doubling condition for pure slabs on the ball is $\Theta\big((\log N/N)^{2/(n+1)}\big)$: both the exponent and the logarithm are exact.

The interpretation is pleasingly sharp. The scaling factor ℓ is the resolution below which the doubling condition does not look; the theorem says that for i.i.d. data this resolution cannot be chosen finer than the scale at which the *most isolated boundary point* of the sample sits alone — and on the ball, among the $\asymp (N/\log N)^{(n-1)/(n+1)}$ essentially independent boundary caps, the most isolated one is isolated to depth $\asymp (\log N/N)^{2/(n+1)}$, a factor $(\log N)^{2/(n+1)}$ deeper than a typical extreme point. The companion upper bound shows this is the only obstruction.

2 Preliminaries

Throughout, $c_4(n) \leq c_5(n)$ denote the spherical cap-volume constants: for the cap $C(u,h) := \{x \in B^n : \langle x,u\rangle > 1-h\}$,

$$c_4\,h^{\frac{n+1}{2}} \;\leq\; \Lambda(h) := \mu\big(C(u,h)\big) \;\leq\; c_5\,h^{\frac{n+1}{2}}, \qquad 0 < h \leq \tfrac{1}{2}, \tag{1}$$

by the standard computation $\Lambda(h) = \frac{1}{\mathrm{vol}(B^n)}\int_{1-h}^{1}\omega_{n-1}(1-t^2)^{\frac{n-1}{2}}\,dt$ and $1-t^2 \asymp (1-t)$ on $[\frac{1}{2},1]$.

We also record the violating-slab mechanism from [4, Theorem 1.3], specialized to pure slabs on the ball.

Lemma 2.1 (Stalled slab)**.** *Fix $\kappa > 1$, $\sigma > 0$, $\delta \geq \frac{3}{N}$. Let u be a unit vector, $0 < h < H \leq \frac{1}{2}$ with $h \leq \frac{H}{4}$, and suppose the sample satisfies: exactly one point $X^* \in C(u,H)$, and moreover $X^* \in C(u,h)$. Then for every $w \leq H/\kappa$, the slab $S_w(u,X^*)$ satisfies*

$$\bar\mu_N\big(S_w(u,X^*)\big) = \bar\mu_N\big(S_{\kappa w}(u,X^*)\big) = \frac{1}{N} \;<\; \min\Big\{\delta,\; (1+\sigma)\,\bar\mu_N\big(S_w(u,X^*)\big)\Big\},$$

i.e. the doubling inequality fails at every scale $w \leq H/\kappa$. Consequently, if some grid scale $\ell\kappa^i$ ($i \in \mathbb{N}_0$, $\ell\kappa^i < \beta$) lies in $(0, H/\kappa]$ — which happens whenever $\ell \leq H/\kappa^2$ and N is large — the UDC with scaling factor ℓ fails.

Proof. Write $\tau(x) = \langle x, u\rangle$ and $\tau^* = \tau(X^*) \geq 1 - h$. For $w' \leq H$,

$$S_{w'}(u, X^*) \cap B^n \;\subseteq\; \{x \in B^n :\; \tau(x) \geq \tau^* - \tfrac{w'}{2}\} \;\subseteq\; C\Big(u,\; h + \tfrac{w'}{2}\Big) \;\subseteq\; C(u, H),$$

since $h + \frac{w'}{2} \leq \frac{H}{4} + \frac{H}{2} < H$. By hypothesis $C(u, H)$ contains only the sample point X^*, which lies in every $S_{w'}(u, X^*)$; hence the empirical masses of S_w and $S_{\kappa w}$ both equal $\frac{1}{N}$ for $w \leq H/\kappa$. Since $\delta \geq \frac{3}{N} > \frac{1}{N}$ and $\sigma > 0$, $\min\{\delta, (1+\sigma)\frac{1}{N}\} > \frac{1}{N}$, and the doubling inequality fails. □

3 Proof of Theorem 1.1

Proof of Theorem 1.1. Fix $\kappa, \sigma, \beta, \delta \geq \frac{3}{N}$ as in the statement, and set

$$q := \big(4\kappa^2\big)^{\frac{n+1}{2}}, \qquad \varepsilon_0 := \frac{n-1}{2(n+1)} \cdot \frac{c_4}{c_5\, q}, \qquad \tau_N := \varepsilon_0 \log N.$$

Define the inner depth h_N by $\Lambda(h_N) = \frac{\tau_N}{N}$ — by (1), $h_N \asymp (\tau_N/N)^{2/(n+1)}$ — and the outer depth $H_N := 4\kappa^2 h_N$, so that by (1),

$$\Lambda(H_N) \;\leq\; c_5 (4\kappa^2)^{\frac{n+1}{2}} h_N^{\frac{n+1}{2}} \;\leq\; \frac{c_5\, q}{c_4} \cdot \frac{\tau_N}{N} \;=:\; \frac{Q\tau_N}{N}. \tag{2}$$

The cap family. The spherical cap $C(u, H_N)$ has angular radius $\arccos(1 - H_N) \leq 2\sqrt{H_N}$. Choose a maximal family of unit vectors $u_1, \dots, u_M$ with pairwise angles at least $8\sqrt{H_N}$; then the caps $C(u_j, H_N)$ are pairwise disjoint, and a standard volumetric packing bound on the sphere gives

$$M \;\geq\; c_6(n)\, H_N^{-\frac{n-1}{2}} \;\geq\; c_7(n, \kappa) \Big(\frac{N}{\tau_N}\Big)^{\frac{n-1}{n+1}}. \tag{3}$$

Per-cap events. Let

$$E_j := \Big\{ \#\big(T \cap C(u_j, H_N)\big) = 1 \;\text{ and }\; T \cap C(u_j, H_N) \subseteq C(u_j, h_N) \Big\},$$

the event that the j-th outer cap contains exactly one sample point and that this point is h_N-deep. Then

$$p := \mathbb{P}(E_j) = N\, \Lambda(h_N) \big(1 - \Lambda(H_N)\big)^{N-1} \;\geq\; \tau_N\, e^{-2Q\tau_N} \;=\; \tau_N\, N^{-2Q\varepsilon_0}, \tag{4}$$

for large N, using $1 - x \geq e^{-2x}$ for $0 \leq x \leq \frac{1}{2}$ and (2).

First moment. Let $X := \sum_{j \leq M} \mathbf{1}_{E_j}$. By (3) and (4),

$$\mathbb{E}X \;=\; Mp \;\geq\; c_7 \Big(\frac{N}{\tau_N}\Big)^{\frac{n-1}{n+1}} \tau_N\, N^{-2Q\varepsilon_0} \;\geq\; c_7\, N^{\frac{n-1}{n+1} - 2Q\varepsilon_0}\, \tau_N^{\,1 - \frac{n-1}{n+1}}.$$

By the choice of ε_0, $2Q\varepsilon_0 = \frac{n-1}{n+1} \cdot \frac{c_4}{c_5} \leq \frac{n-1}{n+1}$, with strict inequality unless $c_4 = c_5$; to be safe replace ε_0 by $\varepsilon_0/2$, so that $2Q\varepsilon_0 \leq \frac{n-1}{2(n+1)}$ and

$$\mathbb{E}X \;\geq\; c_7\, N^{\frac{n-1}{2(n+1)}} \;\longrightarrow\; \infty. \tag{5}$$

Second moment. For $i \neq j$ the caps are disjoint, and

$$\mathbb{P}(E_i \cap E_j) = N(N-1)\, \Lambda(h_N)^2 \big(1 - 2\Lambda(H_N)\big)^{N-2}.$$

Hence

$$\frac{\mathbb{P}(E_i \cap E_j)}{p^2} = \frac{N-1}{N} \cdot \frac{\big(1-2\Lambda(H_N)\big)^{N-2}}{\big(1-\Lambda(H_N)\big)^{2N-2}} \;\le\; \exp\Big(C\,N\,\Lambda(H_N)^2\Big) \;=\; \exp\Big(O\Big(\frac{(\log N)^2}{N}\Big)\Big) \;=\; 1+o(1),$$

uniformly in $i \neq j$, since $\log \frac{(1-2x)}{(1-x)^2} = O(x^2)$ for $0 \le x \le \frac{1}{4}$. Therefore

$$\mathrm{Var}(X) \;=\; \mathbb{E}X + \sum_{i \neq j} \big(\mathbb{P}(E_i \cap E_j) - p^2\big) \;\le\; \mathbb{E}X + o(1)\,(\mathbb{E}X)^2,$$

and by Chebyshev's inequality,

$$\mathbb{P}(X=0) \;\le\; \frac{\mathrm{Var}(X)}{(\mathbb{E}X)^2} \;\le\; \frac{1}{\mathbb{E}X} + o(1) \;=\; o(1)$$

by (5).

Conclusion. With probability $1-o(1)$ some E_j occurs. On E_j, Lemma 2.1 (with $u = u_j$, $h = h_N$, $H = H_N$; note $h_N = H_N/(4\kappa^2) \le H_N/4$) shows the doubling inequality fails at every scale $w \le H_N/\kappa$, so the UDC fails for every scaling factor $\ell \le H_N/\kappa^2 = 4h_N$. Since $h_N \ge (c_5^{-1}\tau_N/N)^{2/(n+1)} \ge c(n,\kappa)(\log N/N)^{2/(n+1)}$, the theorem follows. □

Remark 3.1 (General rolling-ball bodies)**.** *The proof used the ball only through the two-sided cap asymptotic (1) and the packing bound (3). Both hold, with constants depending on the curvature bounds, for any convex body with C^2 boundary and principal curvatures bounded between two positive constants: caps satisfy $\Lambda(h) \asymp h^{(n+1)/2}$ uniformly in the direction, and a maximal separated family of boundary directions provides $\asymp H^{-(n-1)/2}$ disjoint caps. Theorem 1.1 therefore extends verbatim to this class, and with the upper bound of [4] the critical scaling $\Theta((\log N/N)^{2/(n+1)})$ holds for all such bodies. Densities bounded above and below change only the constants.*

Remark 3.2 (The truncated case)**.** *The multiplication-of-configurations mechanism should also upgrade the truncated lower bound of [4] from constant probability at scale $N^{-1/n}$ to probability $1-o(1)$ at scale $(\log N/N)^{1/n}$, matching the upper bound of [3] exactly: since the UDC is quantified over all truncation sets of a given size, one may place $\asymp h_N^{-(n-1)/2}$ rotated copies of a thick conical truncation region with apices along the boundary of a smooth body and run the same second-moment argument over the resulting cone caps, whose masses scale as h^n. The geometric bookkeeping (uniform thickness of the rotated regions and disjointness of the cone caps) is routine but not carried out here.*

Problem 3.3. *Determine the exact constant: does $\ell_N^{\mathrm{crit}}/(\log N/N)^{2/(n+1)}$ converge, and to what, as $N \to \infty$? The upper route loses a factor through the VC bound, the lower route through the choice $H = 4\kappa^2 h$; we expect a clean extremal constant expressible through the cap-volume constant $c_4 = c_4(n)$ and κ alone.*

Acknowledgments

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Israel Science Foundation (grant number 2258/19) and by the Israel Science Foundation (ISF Grant 4101/25).

Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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